

design engineers, product engineers, test engineers, process engineers or applications engineers. Of course, there are other important people like CAD engineers too who keep developing (or maintaining) all the important CAD tools and systems.

Irrespective of which job function you choose, there are certain basic skills required to break into this field. A VLSI engineer is expected to know the physics of semiconductor devices, linear systems, probability and random variables, engineering mathematics (Fourier, Laplace and Z transforms), circuit analysis and engineering electromagnetics. In addition to these solid Verilog/VHDL skills, familiarity with front-end design cycle, synthesis and simulation tool knowledge, sound digital design fundamentals and nonetheless knowledge of microprocessors would be an advantage for any aspiring candidate.

Where to start?

VLSI circuits are everywhere—your personal computer, your cellphone, your brand new state-of-the-art digital camera or for that matter any electronic gadget you dream to buy. Actually, this field involves packing of more and more logic devices into progressively smaller areas. So the circuits that would have taken board-full of space can now be put into a small space few millimetres across!

The best way to start work on VLSI is to know VLSI designing tools. VLSI has been around for a long time, but as a side effect of advances in the world of computers, there has been a dramatic proliferation of tools that can be used to design VLSI circuits. Also, obeying Moore's law, the capability of an IC has increased exponentially over the years in terms of computing power, utilisation of available area and yield. The combined effect of these two factors is that designers can now put diverse functionality into an IC, opening up new frontiers. Examples are embedded systems, where intelligent devices are placed inside everyday objects, and ubiquitous computing, where small

Institute Watch

Advanced Computing Training School (ACTS), Bangalore
Bengal Engineering College (BEC), Howrah
BHU-Institute of Technology (BHU-IT), Varanasi
CEERI, Pilani
Centre for Development of Advanced Computing (C-DAC)
GSITS, Indore
IISc, Bangalore
IITs (Bombay, Delhi, Kanpur, Kharagpur and Chennai)
Jadhavpur University
KREC, Suratkal
Malaviya REC, Jaipur
Manipal Centre for Information Science, Manipal
PSG College of Engineering, Coimbatore
Punjab University, Chandigarh
REC, Rourkela
REC, Warangal
Regional Engineering College, Tiruchirappalli
Roorklee University
Thapar Institute of Engineering and Technology (TIET), Patiala
University of Pune
VRC, Nagpur
Others (ATIIT Pvt Ltd, Bitmapper, Accel Technologies, Trident
Techlabs and Bennis Technologies)

computing devices proliferate to such an extent that even the shoes you wear may actually do something useful like monitoring your heartbeat!

How to deal with VLSI circuits?

Keep in mind that digital VLSI circuits are predominantly CMOS-based. So the way normal blocks like latches and gates are implemented, is different from what you have seen before. However, the behaviour remains the same. All the miniaturisation involves new things to consider. A lot of thought has to go into actual implementation as well as design.

Let us look at some of the external factors responsible for the functionality of the circuit—large complicated circuits running at very high frequencies have to tackle the delays in propagation of signals through gates and wires. This is a big problem even for areas a few micrometres across. The operational speed is so large that as the delays add up, these can actually become comparable to clock speeds.

Another effect of high operation frequencies is increased consump-

tion of power. This has two-fold effect: devices drain batteries faster, and heat dissipation increases. Coupled with the fact that surface areas decrease, heat poses a major threat to the stability of the circuit itself.

Laying out the circuit components is a task common to all branches of electronics. The speciality in VLSI layout is number of possible ways available to do this; there can be multiple layers of different materials on the same silicon, there can be different arrangements of the

smaller parts for the same component and so on.

The power dissipation and speed in a circuit present a trade-off; if you try to optimise on one, the other is affected. The choice between the two is determined by the way you choose to layout the circuit components. Layout can also affect the fabrication of VLSI chips, making it either easy or difficult to implement the components on the silicon.

How to proceed with designs?

Know the fundamentals of both analogue and digital design processes. You may start with a typical design flow of any one of them. For example, a digital design flow to get the end product is: specification→architecture→RTL coding→RTL verification→synthesis→backend→tape out to foundry. Here the end product is a wafer with repeated number of identical ICs.

Today, all digital designs start with a designer writing a hardware description of the IC (using hardware description language, or HDL) in Verilog/VHDL. A Verilog or VHDL program essentially describes the